

Switching-Cost-Aware Deep Reinforcement Learning for Dynamic Port Selection in Fluid Antenna Systems

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Abstract—Fluid antenna systems (FAS) have emerged as a promising enabling technology for sixth-generation (6G) wireless networks. By dynamically switching ports according to channel characteristics, FAS can exploit spatial diversity and significantly enhance wireless performance. However, most existing studies focus on maximizing instantaneous throughput while neglecting the long-term cost associated with frequent port switching. This work aims to jointly optimize long-term throughput and port-switching cost in FAS, considering spatio-temporally correlated channels and minimum-rate requirements for each user. To address this challenge, we formulate the dynamic port selection problem as a Markov decision process (MDP) and employ a dueling DQN framework to obtain a near-optimal policy. A Transformer encoder is incorporated to capture spatial-temporal dependencies among ports, and a candidate pre-selection mechanism is introduced to reduce training complexity. Simulation results show that the proposed method achieves up to 96% of the per-slot exhaustive-search optimum while significantly reducing decision latency, demonstrating high efficiency and scalability for large-scale FAS deployments.

Index Terms—Fluid antenna system (FAS), port selection, deep reinforcement learning (DRL), switching cost.

I. INTRODUCTION

ORIGINALLY introduced by Wong et al. in [1] and [2], the fluid antenna system (FAS) concept represents a novel paradigm of treating antenna as a reconfigurable physical-layer resource emphasizing shape and position flexibility in antenna to empower the physical layer [3], [4], [5]. Practical fluid antennas might take many different forms, from liquid-based structures [6] to metasurfaces [7], [8] and to pixel-based designs [9], etc. Position reconfigurability offers a new way to obtain spatial diversity for reliability and capacity benefits [10], as well as multiple access [11], [12].

In FAS, the ability to determine the best position (i.e., the port selection problem) dictates the system performance that also depends upon the availability of channel state information

(CSI) [13]. Traditional methods of port selection typically rely on heuristic or learning-based algorithms. For instance, [14] studied the use of cross entropy optimization (CEO) for joint port selection of fluid antennas at both ends. Machine learning methods such as [15] also appear to be effective in FAS port selection, and a recent work [16] further proposed a large language model (LLM)-based intelligent framework for jointly optimizing port selection and beamforming in multiuser multiple-input multiple-output (MIMO)-FAS systems.

Deep learning methods have recently emerged as a promising approach to port selection in FAS [17]. By training neural networks to directly map channel states to optimal port configurations, these methods eliminate the need for explicit modeling. For instance, the work presented in [18] employed an online learning scheme to track dynamic channel conditions. These learning-based strategies enhance computational efficiency and maintain adaptability to dynamic channel conditions.

Unlike typical deep learning, deep reinforcement learning (DRL) continuously interacts with the environment to optimize long-term rewards by considering state transitions. This makes DRL particularly well-suited for dynamic decision problems characterized by temporal correlations and action dependencies. For instance, the authors of [19] introduced multi-agent DRL for multiuser port selection without requiring other users' CSI, whereas [20] combined DRL with a pointer network to jointly optimize port selection and beamforming.

However, existing DRL-based FAS schemes predominantly focus on maximizing instantaneous throughput or minimizing channel estimation overhead. They largely overlook the physical switching energy cost inherent to high-density fluid antennas. Although switching-related trade-offs have been examined in broader wireless resource allocation literature, the unique characteristics of FAS render physical switching energy a critical factor for long-term system sustainability. In particular, FAS features high port density and strong spatio-temporal channel correlations. Ignoring this cost can lead to excessive port toggling, resulting in unsustainable hardware energy consumption and increased control latency [21].

Motivated by this, we propose a DRL framework designed to effectively balance the trade-off between instantaneous throughput and long-term switching overhead. Our solution incorporates a Transformer-based architecture to capture complex spatial-temporal correlations in FAS channels, enabling more informed and stable decision-making. Additionally, we introduce an efficient candidate pre-selection mechanism to enhance training tractability for practical deployment. Specifically, our main contributions are summarized as follows:

- First, we model the port selection problem as a weighted combination of instantaneous throughput and physical

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switching energy cost. Considering the spatial correlations and temporal dynamics of channel states, we transform this optimization problem into a Markov decision process (MDP) and employ a dueling deep Q-network (DQN) framework to learn near-optimal long-term port selection policies.

- Based on the dueling DQN framework, we further develop an enhanced architecture that integrates a candidate pre-selection mechanism and a Transformer-based state encoder to improve sampling efficiency and spatial-temporal modeling capability. Simulation results demonstrate that the proposed method maintains robust performance under number of ports, achieves up to 96% of the per-slot exhaustive search throughput, and significantly reduces execution time compared with exhaustive search.

II. SYSTEM MODEL AND PROBLEM FORMULATION

We consider a multiuser FAS system, in which the base station (BS) is equipped with a two-dimensional (2D) fluid antenna to serve K single-antenna users. The FAS at the BS consists of $N = N_x \times N_y$ candidate ports, which are uniformly distributed over a 2D surface with an area $W = W_x \lambda \times W_y \lambda$, where λ is the carrier wavelength. The N_i ports are evenly distributed along a straight line of length $W_i \lambda$ for $i \in \{x, y\}$. In this multiuser FAS system, the BS activates only M out of N ports to serve the K users, with $N \geq M \geq K$.

At time slot t , the BS activates a subset of ports, denoted by $\mathcal{S}_t \subseteq \{1, \dots, N\}$ with $|\mathcal{S}_t| = M$, to transmit data to the users. Let $\mathbf{h}_{k,\mathcal{S}_t}(t) \in \mathbb{C}^M$ denote the channel vector from the selected ports $\mathcal{S}_t$ to user k , and let $\mathbf{W}(t) = [\mathbf{w}_1(t), \dots, \mathbf{w}_K(t)] \in \mathbb{C}^{M \times K}$ denote the corresponding linear precoding matrix. The received signal at user k is expressed as

$$y_k(t) = \mathbf{h}_{k,\mathcal{S}_t}^H(t) \mathbf{w}_k(t) x_k(t) + \sum_{j \neq k} \mathbf{h}_{k,\mathcal{S}_t}^H(t) \mathbf{w}_j(t) x_j(t) + n_k(t), \quad (1)$$

where $x_k(t)$ is the transmitted symbol with $\mathbb{E}[|x_k(t)|^2] = 1$, and $n_k(t) \sim \mathcal{CN}(0, \sigma^2)$ is additive white Gaussian noise.

In a rich-scattering environment, according to the Jakes' model [10], the spatial correlation between the (n_x, n_y) -th and $(\tilde{n}_x, \tilde{n}_y)$ -th ports in FAS is characterized by

$$J_{(n_x, n_y), (\tilde{n}_x, \tilde{n}_y)} = J_0 \left(2\pi \sqrt{\left(\frac{|n_x - \tilde{n}_x| W_x}{N_x - 1} \right)^2 + \left(\frac{|n_y - \tilde{n}_y| W_y}{N_y - 1} \right)^2} \right), \quad (2)$$

where $J_0(\cdot)$ is the zero-order Bessel function of the first kind. The 2D indices are mapped into a 1D index as $m = (n_y - 1)N_x + n_x$, yielding the spatial correlation matrix

$$\mathbf{R} = [J_{m,m'}]_{N \times N}.$$

Temporal correlation caused by user mobility follows a first-order autoregressive (AR (1)) model [18]:

$$\mathbf{h}_k(t) = \rho \mathbf{h}_k(t-1) + \sqrt{1 - \rho^2} \mathbf{e}_k(t), \quad (3)$$

where $\rho = J_0(2\pi f_D T_s)$ denotes the temporal correlation coefficient, f_D is the Doppler frequency, T_s is the slot duration,

and $\mathbf{e}_k(t) \sim \mathcal{CN}(\mathbf{0}, \beta_k \mathbf{R})$ is the innovation term with large-scale fading factor $\beta_k > 0$. To ensure the AR (1) process is wide-sense stationary, the initial channel vector is set to follow the steady-state distribution $\mathbf{h}_k(0) \sim \mathcal{CN}(\mathbf{0}, \beta_k \mathbf{R})$.

From (1), the signal-to-interference-plus-noise ratio (SINR) of user k under the selected port set $\mathcal{S}_t$ is given by

$$\text{SINR}_k(t) = \frac{|\mathbf{h}_{k,\mathcal{S}_t}^H(t) \mathbf{w}_k(t)|^2}{\sum_{j \neq k} |\mathbf{h}_{k,\mathcal{S}_t}^H(t) \mathbf{w}_j(t)|^2 + \sigma^2}. \quad (4)$$

When inter-user interference (IUI) is suppressed (e.g., by zero forcing (ZF) or orthogonal scheduling), the received signal-to-noise ratio (SNR) and the rate are simplified to

$$\begin{cases} \text{SNR}_k(t) = \frac{|\mathbf{h}_{k,\mathcal{S}_t}^H(t) \mathbf{w}_k(t)|^2}{\sigma^2}, \\ R_k(t) = \log_2(1 + \text{SNR}_k(t)). \end{cases} \quad (5)$$

Since this work focuses on the port-selection mechanism rather than beamforming design, ZF precoding is adopted to eliminate IUI although other linear precoding schemes can also be employed without affecting the generality of the analysis.

Frequent port switching can lead to excessive control and hardware overhead. To study this, we define the number of ports whose states change between two consecutive slots as

$$C_t = \text{card}(\mathcal{S}_t \triangle \mathcal{S}_{t-1}), \quad (6)$$

where $\triangle$ is the symmetric difference operator and $\text{card}(\cdot)$ represents the cardinality of a set. Let $e_{\text{sw}} > 0$ denote the energy consumed for switching one port once. Then the total switching energy at slot t is $E_{\text{sw}}(t) = e_{\text{sw}} C_t$. This linear model assumes that switching energy is proportional to the number of switching events, neglecting second-order effects such as thermal accumulation or device aging.

Considering both achievable rate and switching overhead, the long-term optimization over T slots is formulated as

$$\max_{\{\mathcal{S}_t\}} \sum_{t=1}^T \left[\sum_{k=1}^K R_k(t) - \eta_{\text{sw}} E_{\text{sw}}(t) \right], \quad (7a)$$

$$\text{s.t. } \mathcal{S}_t \subseteq \{1, \dots, N\}, \quad \forall t, \quad (7b)$$

$$|\mathcal{S}_t| = M, \quad \forall t, \quad (7c)$$

$$R_k(t) \geq R_{\min}, \quad \forall k, t. \quad (7d)$$

In (7a), $\eta_{\text{sw}} > 0$ balances rate and switching energy. Note that $E_{\text{sw}}(t)$ is the physical switching energy of port reconfiguration, not the transmit power. The weighted-sum form enables a tunable trade-off between throughput and switching cost. Constraints (7b) & (7c) define the feasible port-selection space, while (7d) enforces per-user quality-of-service requirements. Problem (7) constitutes a temporally coupled combinatorial optimization task due to the inter-slot dependency by (6). A DRL-based port-selection framework is developed in the next section to efficiently learn adaptive scheduling policies under spatial-temporally correlated FAS channels.

III. DRL-BASED PORT SELECTION MODEL

At time slot t , we have the multiuser channel matrix

$$\mathbf{H}_t = [h_{k,n}(t)] \in \mathbb{C}^{K \times N}, \quad (8)$$

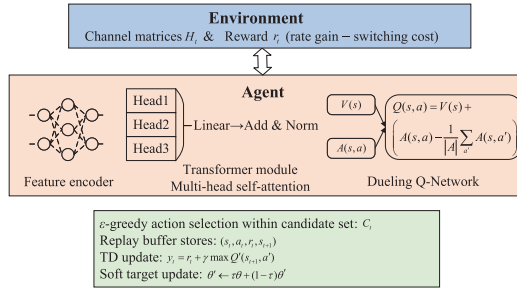


Fig. 1. The proposed framework for dynamic port selection in FAS.

in which $h_{k,n}(t)$ represents the complex channel coefficient between user k and FAS port n . The BS (agent) observes the current channel environment and determines the optimal subset of active ports to maximize the long-term performance objective in (7). We formulate the port-selection process as an MDP $\mathcal{M} = \langle \mathbb{S}, \mathcal{A}, R \rangle$, where $\mathbb{S}$ denotes the DRL state space (to avoid confusion with the port set $\mathcal{S}_t$).

A. State

The state $s_t = [\mathbf{H}t, \mathbf{o}t - 1, C_{t-1}]$ captures the current channel condition $\mathbf{H}t$, the previous active port indicator $\mathbf{o}t - 1 \in 0, 1^N$, and the number of port switches C_{t-1} between slots $t - 2$ and $t - 1$.

B. Action

Each discrete action a_t corresponds to selecting an M -port subset $\mathcal{S}_t = \{n_1, \dots, n_M\} \subseteq \{1, \dots, N\}$. The action space therefore contains $\binom{N}{M}$ combinations. To reduce complexity, we introduce a candidate port set $\mathcal{C}_t$ and restrict actions to the reduced domain $\mathcal{A}_t \triangleq \{a \subseteq \mathcal{C}_t \mid |a| = M\}$, so that actions are chosen from $\binom{|\mathcal{C}_t|}{M}$ candidates.

C. Reward

To ensure consistency with the optimization objective, the per-slot reward is defined as

$$r_t = \sum_{k=1}^K R_k(t) - \eta_{\text{sw}} E_{\text{sw}}(t), \quad (9)$$

which balances the sum-rate and the port-switching cost.

Although the state, action, and reward of the MDP are well defined, the combinatorial nature of the action space makes conventional DRL algorithms difficult to apply directly. To address this challenge, we design a hierarchical optimization framework, as illustrated in Fig. 1, which integrates candidate pre-selection, Transformer encoder, and local refinement.

1) *Candidate Pre-Selection*: To mitigate the combinatorial explosion, a heuristic scoring mechanism is introduced to pre-select promising ports before the DRL decision stage. The score for port n at slot t is obtained by

$$\gamma_n = \|\mathbf{h}_n(t)\|_2^2 + \eta_{\text{sw}} o_{t-1,n}, \quad (10)$$

where $\mathbf{h}_n(t) = [h_{1,n}(t), h_{2,n}(t), \dots, h_{K,n}(t)]^T$ represents the aggregated channel power of port n across all users. The top- L ports form the reduced candidate set $\mathcal{P}_t$, reducing the action space from $\binom{N}{M}$ to $\binom{L}{M}$. Such dimensionality reduction increases the probability of including high-quality ports in the

candidate set, while the DQN's exploration mechanism (e.g., ϵ -greedy) ensures sufficient action diversity within $\mathcal{P}_t$ to avoid suboptimal convergence.

2) *Transformer-Assisted Dueling DQN*: On top of a DQN, a Transformer encoder and a dueling head are used to improve spatial-temporal feature extraction and value estimation. We adopt a lightweight Transformer architecture over RNN-based (e.g., LSTM) or CNN alternatives. The self-attention mechanism reduces sensitivity to scenario-specific hyperparameters by data-dependently identifying relevant features without predefined receptive fields, while the parallel structure enables low-latency inference. The Transformer processes $\mathbf{H}_t$ to model inter-port correlation and temporal dynamics, allowing the agent to exploit contextual information beyond CSI. The dueling structure separately estimates the state value $V(s_t)$ and advantage $A(s_t, a_t)$:

$$Q(s_t, a_t; \theta) = V(s_t; \theta_v) + \left[A(s_t, a_t; \theta_a) - \frac{1}{|\mathcal{A}_t|} \sum_{a' \in \mathcal{A}_t} A(s_t, a'; \theta_a) \right], \quad (11)$$

where the normalization ensures unbiased Q-value estimation. This hybrid design captures complex spatial-temporal dependencies and improves robustness against correlated inputs. Compared with standard DQN, it provides richer state abstraction and smoother convergence, for time-varying channels.

3) *TD Target and Mini-Batch Training*: The agent stores transitions (s_t, a_t, r_t, s_{t+1}) in a replay buffer $\mathcal{D}$, and mini-batches of size B are randomly sampled to decorrelate updates. The temporal-difference (TD) target and loss are

$$y_t = r_t + \gamma \max_{a' \in \mathcal{A}_{t+1}} Q_{\theta^-}(s_{t+1}, a'), \quad (12)$$

$$\mathcal{L}_{\text{batch}} = \frac{1}{B} \sum_{b=1}^B (y_b - Q_{\theta}(s_b, a_b))^2. \quad (13)$$

Network parameters are updated via gradient descent and soft target replacement:

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}_{\text{batch}}; \quad \theta^- \leftarrow \tau \theta + (1 - \tau) \theta^-. \quad (14)$$

This iterative learning process refines the policy while avoiding divergence under correlated inputs.

4) *Local Refinement*: Although the DRL agent produces high-quality actions, its output $\hat{\mathcal{S}}_t$ may still be locally suboptimal. To further improve performance, a lightweight one-swap local search is applied at inference. Starting from $\hat{\mathcal{S}}_t$, one port is iteratively replaced if it increases the reward in (9), while maintaining $|\mathcal{S}_t| = M$, until convergence. This refinement bridges the small gap between the DRL output and the exhaustive optimum, with linear complexity $\mathcal{O}(TN)$ per episode, far lower than $\mathcal{O}\left(T \binom{N}{M}\right)$ for brute-force enumeration.

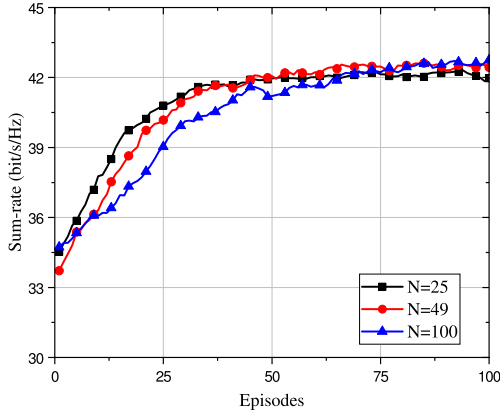
The complete training and inference procedure is summarized in Algorithm 1, which trains the Transformer-assisted dueling DQN to maximize the long-term objective (7).

IV. SIMULATION RESULTS

Simulations are conducted in Python (PyTorch 2.2) on a workstation with an AMD Ryzen 9 7940HX CPU, 32 GB RAM, and an NVIDIA RTX 4070 GPU. Unless otherwise specified, the FAS comprises a 5×5 grid of candidate ports

Algorithm 1 Training Procedure of the Proposed Algorithm

- 1: Initialize Transformer-assisted dueling DQN Q_θ , target $Q_{\theta^-} \leftarrow Q_\theta$, and replay buffer $\mathcal{D}$.
- 2: **for** each episode **do**
- 3: **for** each slot t **do**
- 4: Observe state s_t and channel $\mathbf{H}_t$.
- 5: Compute scores $\gamma_n = \|\mathbf{h}_n(t)\|_2^2 + \omega_{\text{sw}} o_{t-1,n}$.
- 6: Select top- L ports as $\mathcal{P}_t$, construct $\mathcal{A}_t = \{a \subseteq \mathcal{P}_t : |a| = M\}$.
- 7: Choose a_t via ε -greedy policy, execute it, obtain r_t , observe s_{t+1} , and store (s_t, a_t, r_t, s_{t+1}) in $\mathcal{D}$.
- 8: Sample mini-batch $\{(s_b, a_b, r_b, s'_b)\}$; compute $y_b = r_b + \gamma \max_{a'} Q_{\theta^-}(s'_b, a')$.
- 9: Update θ by minimizing $\frac{1}{B} \sum_b (y_b - Q_\theta(s_b, a_b))^2$ and softly update $\theta^- \leftarrow \tau\theta + (1 - \tau)\theta^-$.
- 10: **end for**
- 11: **end for**

Fig. 2. Convergence under different numbers of candidate ports N .

($N = 25$), among which $M = 5$ are activated per slot to serve $K = 3$ single-antenna users. The noise power is $\sigma^2 = 10^{-3}$, and channels follow Jakes' model with spatial and temporal correlation over $T = 100$ slots. The per-switch energy cost is normalized to $e_{\text{sw}} = 1$.

The proposed algorithm is built on an improved dueling DQN with four fully connected layers (128, 256, 256, 128) and a two-layer Transformer encoder (3 heads, hidden size 128). We use the Adam optimizer (learning rate 10^{-3}), discount factor $\gamma = 0.95$, replay buffer of 10^5 , mini-batch size 64, and soft target update $\tau = 0.01$. An ε -greedy strategy decays from 1.0 to 0.05. Results are averaged over five runs.

We first evaluate convergence and scalability under different system dimensions with $N = 25, 49$, and 100, while keeping $M = 5$ and $K = 3$. As shown in Fig. 2, the algorithm converges stably within roughly 50 episodes in all cases. Larger N leads to slightly slower convergence and more early oscillations due to increased exploration difficulty, while final performance improves moderately with more ports due to enhanced spatial diversity. These results verify the scalability of the proposed method across different system sizes.

In Fig. 3, we compare the proposed method with four baselines: exhaustive search (per-slot), random selection, standard DQN, and proximal policy optimization (PPO). The exhaustive search baseline enumerates all possible port

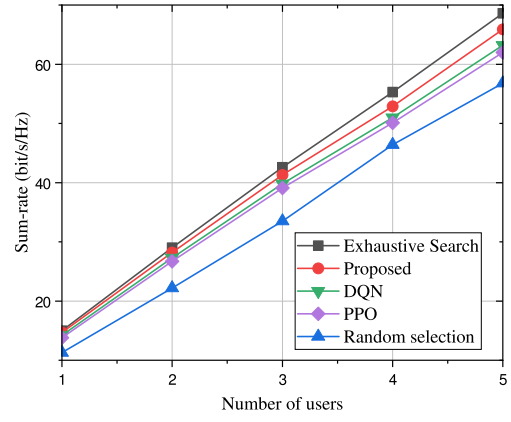


Fig. 3. Sum rate versus number of users.

TABLE I
CONTRIBUTION OF EACH ARCHITECTURAL COMPONENT

Method	Conv. (eps)	Sum-Rate (bit/s/Hz)	Gain
DQN	40	39.97	—
DQN + Pre-selection	26	41.54	+3.9%
DQN + Transformer	52	41.81	+4.6%
DQN + Local refinement	43	40.43	+1.2%
Full proposed method	38	42.10	+5.3%

combinations within each time slot to maximize the instantaneous sum-rate, serving as a per-slot performance upper bound. The random selection baseline randomly chooses a feasible port subset from the pre-filtered candidate set $\mathcal{P}_t$ (i.e., ports with high instantaneous gains, satisfying $|\mathcal{S}_t| = M$). DQN and PPO adopt the same network structure as the proposed model for fair comparison. Due to the exponential complexity of exhaustive search, experiments are conducted under the 25-choose-5 configuration. We clarify that this exhaustive search is performed independently within each time slot rather than over the entire T -slot trajectory. A global search would require enumerating $\binom{25}{5}^T$ possible sequences and is computationally infeasible. In this experiment, the switching weight is set to $\eta_{\text{sw}} = 0.1$, making the optimization objective in Eq. (7) approximate sum-rate maximization. Therefore, this per-slot optimum serves as a valid reference for evaluating the core port selection performance of our DRL agent. As the number of users increases, the performance of DRL-based algorithms decreases slightly due to the much enlarged state space, but the proposed method achieves the best performance-complexity trade-off, reaching up to 96% of the per-slot exhaustive-search optimum with significantly lower computational cost. To quantify the contribution of each component, we conduct ablation studies with results summarized in Table I. Pre-selection substantially accelerates convergence, reducing required episodes from 40 to 26. The Transformer encoder yields the largest performance gain (+4.6%), confirming its effectiveness in capturing spatio-temporal dependencies. Local refinement provides a modest but consistent improvement in final performance ($\approx 1\%$ sum-rate gain) with negligible impact on overall inference time.

The effect of the switching-penalty coefficient η_{sw} is shown in Fig. 4, where $\eta_{\text{sw}} \in 0.1, 1, 5, 10, 20$. At $\eta_{\text{sw}} = 20$, the rate gain and switching penalty are comparable in magnitude, making their trade-off clearly observable. As η_{sw} increases, the number of port changes decreases, improving scheduling

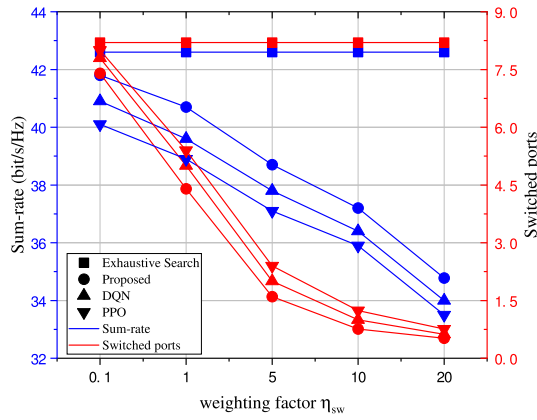


Fig. 4. Sum rate and number of switched ports versus weight factor η_{sw} .

TABLE II

TRAINING AND INFERENCE TIME COMPARISON OF DIFFERENT METHODS

Method	Training Time (min)	Execution Time (ms/slot)
Exhaustive Search	–	841.7
Random (in $\mathcal{P}_t$)	–	0.21
DQN	16	0.38
Proposed	42	0.87
PPO	71	0.43

stability, while the average throughput drops moderately due to more conservative switching. When $\eta_{sw} = 5$, the system achieves a good balance between rate performance and stability, demonstrating that the proposed model can flexibly adapt to different service requirements by tuning a single parameter. Note that the optimal η_{sw} depends on the operating SNR regime, since the reward magnitude scales with the achievable rate while the switching penalty remains fixed. For different deployment scenarios, this parameter can be calibrated offline to maintain the desired performance-stability trade-off.

Finally, Table II summarizes the runtime comparison. Exhaustive search requires about 841.7 ms per slot, while random selection runs almost instantaneously (0.21 ms/slot) but yields sub-optimal results. The standard DQN achieves 0.38 ms/slot inference latency. The proposed model increases this to 0.7–0.9 ms/slot due to attention-based feature extraction, yet remains nearly three orders of magnitude faster than exhaustive search (970× speed-up) while delivering near-optimal accuracy. PPO exhibits comparable inference latency (0.4–0.45 ms) but longer training time owing to on-policy sampling. Overall, the proposed method achieves the best balance among accuracy, stability, and computational efficiency.

V. CONCLUSION

In this letter, we proposed a deep reinforcement learning (DRL)-based adaptive port selection framework for fluid antenna systems (FAS). By integrating a Transformer encoder with a dueling Q-network architecture, the proposed method effectively captures spatio-temporal channel correlations while maintaining low computational complexity. Simulation results demonstrate that the algorithm achieves near-optimal performance, up to 96% of the exhaustive-search benchmark, while significantly reducing computation time, and provides flexible control of the throughput stability trade-off through a single switching-weight parameter. Compared with conventional DQN and PPO baselines, the proposed approach

achieves higher accuracy and more stable decisions at the cost of a slightly increased inference time. Future work will extend this framework to multi-cell cooperative scenarios and investigate joint beam selection and power control for large-scale FAS deployments.

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